Infrared Small Moving Target Detection via Motion Constrained Robust Principal Component Analysis

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***Abstract*—The rapid growth of the low-altitude economy has generated heightened research interest in low-altitude target detection. We focus on the challenging problem of detecting small moving targets that are often diminutive in size and may blend into the background. This paper proposes a motion-constrained robust principal component analysis (MC-RPCA) framework to achieve effective small moving target detection. Specifically, MC-RPCA introduces a motion-derived weight map computed from optical flow with global-motion compensation to downweight dynamic-background residuals, as well as a trajectory-confidence map, which enforces temporal continuity and suppresses impulsive noise. The resulting per-frame convex problem is efficiently solved via an ADMM-based algorithm, enabling streaming processing. Experimental results demonstrate that MC-RPCA achieves substantially fewer false alarms under dynamic backgrounds, while maintaining a low miss rate, while maintaining a low miss rate compared with classic RPCA and YOLOv10n, and yields more stable and accurate detection.**

***Index Terms*—robust principal component analysis, optical flow, infrared small moving target detection, motion constraints**

I. Introduction

The rapid growth of the low-altitude economy [1]–[3] has led to increasing demands for reliable sensing of low-altitude flight activities using radar, vision, and infrared (IR) imaging. Significant research efforts have been made in low-altitude target detection, ranging from target classification [4] to tracklet association [5], [6]. Most existing work focuses on the detection and recognition of standard targets, while research on detecting small moving targets remains less explored. Small moving targets typically occupy only a few pixels and move within the field of view, resulting in a low signal-to-noise ratio as captured by sensors [7]–[9]. In this context, IR imaging offers a viable solution for small moving target detection, as it is less sensitive to variations in illumination and performs robustly at night or in adverse weather conditions [10], [11]. However, IR detection of small moving targets continues to be a challenging task within the realm of low-altitude sensing.

In the literature, there are two primary approaches for target detection. One category includes deep learning-based one-stage detectors, such as the YOLO family, known for their speed and end-to-end training [12]–[15]. However, the study of YOLO for IR small moving target detection is still limited, primarily due to the scarcity of high-quality IR imagery and the limited discriminative features of the targets, leading to missed and false alarms [16]. Alternatively, sparse-and-low-rank decomposition methods model the background as a low-rank component and recognize targets as sparse outliers. Robust Principal Component Analysis (RPCA) offers a principled unsupervised framework for this separation and has been applied to IR small target detection [17]–[19]. However, classic RPCA overlooks inter-frame motion and temporal continuity, making it susceptible to dynamic backgrounds and noise. Consequently, its detection performance can be unstable and prone to false alarms [20].

In this paper, we tackle the challenge of IR small moving target detection. We incorporate target motion information into a reweighted RPCA formulation, leveraging the temporal continuity of small moving targets to enhance detection sensitivity. Additionally, the motion constraint helps mitigate the effects of random impulse noise and background interference. To achieve this, we employ an optical flow method to construct a pixel-wise weighting matrix that emphasizes the moving target while diminishing the influence of random noise and background interference. Furthermore, we model motion prediction based on previous frames to establish a confidence range for the

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moving target in the current frame. These two weighting matrices are crucial for capturing the motion information of small moving targets. At each frame, the detection problem is formulated as a reweighted RPCA problem, for which we propose an efficient Alternating Direction Method of Multipliers (ADMM) algorithm. Our method is unsupervised and allows for streaming processing across frames. Through simulations, our proposed algorithm demonstrates promising detection accuracy and a significantly lower false alarm rate compared to YOLO and classic RPCA.

II. Problem Statement

A. Infrared (IR) Small Target Detection

Low-altitude IR small target detection aims to localize small and dim targets embedded in complex background clutter. In this work, we consider an IR image sequence $\{\boldsymbol{I}_t\}_{t=1}^{T}$, where each frame $\boldsymbol{I}_t \in \mathbb{R}^{m\times n}$. Our goal is to detect and localize targets by separating each observed frame into a background component and a target (foreground) component, which can be formulated as:

$$\boldsymbol{I}_t = \boldsymbol{L}_t + \boldsymbol{S}_t, \tag{1}$$

where $\boldsymbol{L}_t$ denotes the background and $\boldsymbol{S}_t$ denotes the target response to be estimated.

Low-altitude IR perception faces significant challenges due to the following target characteristics: (1) targets are often small, typically occupying only a few pixels, which provides limited shape and texture cues for reliable localization; (2) targets commonly exhibit a low signal-to-noise ratio (SNR) and can be easily submerged in background clutter and sensor noise, resulting in weak and ambiguous target responses; and (3) target visibility may be intermittent due to occlusion and complex scene dynamics, which hampers stable observation and leads to temporally inconsistent detections.

B. Limitations of Existing Methods

Existing IR small target detection methods can be broadly grouped into decomposition-based approaches and learning-based detectors. A representative classic decomposition model is Robust Principal Component Analysis (RPCA), which separates an observed frame $\boldsymbol{I}_t \in \mathbb{R}^{m\times n}$ into a low-rank background $\boldsymbol{L}_t$ and a sparse foreground $\boldsymbol{S}_t$ by solving:

$$\begin{aligned} \min_{\boldsymbol{L}_t,\boldsymbol{S}_t} \quad & \|\boldsymbol{L}_t\|_* + \lambda\|\boldsymbol{S}_t\|_1 \\ \text{s.t.} \quad & \boldsymbol{I}_t = \boldsymbol{L}_t + \boldsymbol{S}_t, \end{aligned} \tag{2}$$

where $\|\cdot\|_*$ is the nuclear norm, $\|\cdot\|_1$ is the element-wise ℓ_1 norm, and $\lambda > 0$ is a given penalty parameter. RPCA is typically applied in a frame-wise manner and therefore does not explicitly encode motion cues or temporal continuity. Moreover, due to the target being small and dim, RPCA may make mistakes by absorbing abnormal patterns, including platform jitter, parallax, impulsive noise, and background residuals, into the sparse term. This could mix the undesired signals with the moving target and increase the false alarm rate.

More recently, deep learning approaches have gained increasing research interest, such as the YOLO family [12]–[15]. On the other hand, such deep learning methods require a massive scale of high-quality annotations as a prerequisite, but the IR small target scenario lacks data and professional annotations. In this context, YOLO is not specifically designed for infrared small moving-target detection, as it does not account for targets' small size and low SNR. As will be demonstrated by simulations, YOLO may often fail to detect the IR small moving target.

III. Proposed Problem Formulation

A. Motion-Constrained RPCA (MC-RPCA) Model

In order to enhance the robustness of detecting small, dim, and moving targets in IR images, we propose to embed the motion dynamics into the RPCA framework, leading to a new and adaptive weighted RPCA.

In particular, we leverage the optical flow method across multiple frames, which incorporates motion compensation into the sparse modeling. Then, pixels with larger residual motion are more likely to correspond to moving targets, whereas pixels with small residual motion are more likely to be the background. Accordingly, we construct a motion-derived nonnegative weight $\boldsymbol{W}_t \in \mathbb{R}^{m\times n}$ from optical flow and use it to reweight the sparsity penalty, and a small value of $\boldsymbol{W}_t$ will promote the moving target to be detected.

Moreover, we utilize the temporally continuous trajectories of the target explicitly for predicting motions among frames. So we further construct a trajectory-confidence map $\boldsymbol{M}_t \in [0,1]^{m\times n}$ by propagating the previous sparse estimate to the current frame using the estimated flow. The confidence map $\boldsymbol{M}_t$ provides a regularization on the trajectory continuity, and helps distinguish targets from misleading isolated interference.

As a result, we arrive at the following motion-constrained RPCA formulation for IR small moving target detection:

$$\begin{aligned} \min_{\boldsymbol{L}_t,\boldsymbol{S}_t} \quad & \|\boldsymbol{L}_t\|_* + \lambda\|\boldsymbol{W}_t \odot \boldsymbol{S}_t\|_1 + \gamma\|(1-\boldsymbol{M}_t)\odot \boldsymbol{S}_t\|_1 \\ \text{s.t.} \quad & \boldsymbol{I}_t = \boldsymbol{L}_t + \boldsymbol{S}_t, \end{aligned} \tag{3}$$

where $\odot$ denotes Hadamard product, $\boldsymbol{W}_t$ is a *motion-derived* weight map, and $\boldsymbol{M}_t$ is a *trajectory-confidence* map. Parameters $\lambda > 0$ and $\gamma > 0$ control the strengths of motion-weighted sparsity and trajectory continuity, respectively.

B. Construction of Motion and Trajectory Priors

Next, we introduce the construction of the weight matrices $\boldsymbol{W}_t$ and $\boldsymbol{M}_t$.

Motion-derived weight map $\boldsymbol{W}_t$: The weight map $\boldsymbol{W}_t$ accounts for the sparsity penalty induced by the apparent motion between two consecutive IR frames $\boldsymbol{I}_{t-1}$ and $\boldsymbol{I}_t$. In order to obtain $\boldsymbol{W}_t$, we first need to compute a velocity field $\boldsymbol{V}_t$. Due to camera-motion-induced disturbances, we propose to estimate $\boldsymbol{V}_t$ via an optical flow method compensated by median removal. To be specific, we take the Lucas-Kanade [21] algorithm as the optical flow method. Without explicit camera-motion priors, an engineering-friendly approach is to

subtract the median of the velocity field from $\boldsymbol{V}_t$, which largely cancels the global motion component and mitigates camera-motion-induced disturbances.

We design the following motion-aware weights based on the motion-compensated velocity field $\boldsymbol{V}_t^{com} \in \mathbb{R}^{m\times n}$,

$$\boldsymbol{W}_t(x,y) = w_l + (w_h - w_l)\cdot \sigma\left(\frac{\tau - \|\boldsymbol{V}^{com}(x,y)\|_2}{k}\right) \quad (4)$$

$$\sigma(z) = \frac{1}{1+\exp(-z)} \quad (5)$$

where τ is a threshold, k controls smoothness, and w_l and w_h respectively denote the low and high weights.

Trajectory-confidence map $\boldsymbol{M}_t$: The role of the weight map $\boldsymbol{M}_t$ is to impose a regularization on the target trajectory continuity, which can eliminate the influence of impulsive noise. Therefore, the weight map $\boldsymbol{M}_t$ contains the trajectory-confidence information that predicts a likely target region in frame t based on historical sparse estimates.

We can produce a prediction $P_t \in \mathbb{R}^{m\times n}$ of the target S_t in the frame t, by leveraging the detection result from the previous frame S_{t-1} and the optical flow computed from frame $t-1$ to frame t,

$$P_t = T_{t-1\to t}(\boldsymbol{S}_{t-1}), \quad (6)$$

where $T_{t-1\to t}(\cdot)$ denotes prediction operator induced by the optical flow from frame $t-1$ to frame t. We design a confidence map M_t based on the distance to the predicted target location:

$$\boldsymbol{M}_t(x,y) = \exp\left(-\frac{d\big((x,y),\ \mathrm{supp}(P_t)\big)^2}{2\sigma_m^2}\right), \quad (7)$$

where $\mathrm{supp}(\cdot)$ denotes the support set and σ_m controls the decay scale. A larger value of M_t indicates a higher probability that the pixels in the corresponding region belong to the target.

IV. Optimization Algorithm

In this section, we custom build an optimization method for the resulting problem (3), which is specifically designed for a frame-continuous IR small target detection setting. At time t, once a new frame $\boldsymbol{I}_t$ is acquired, we first compute the motion-derived weight map $\boldsymbol{W}_t$ from optical flow and construct the trajectory-confidence map $\boldsymbol{M}_t$ from the previous sparse estimate along the estimated flow. With $(\boldsymbol{W}_t, \boldsymbol{M}_t)$ fixed, the problem (3) becomes a low-rank plus reweighted-sparse optimization problem on the *current* frame, and the problem is convex.

Due to the nonnegativity of W_t and $(1-M_t)$, the two weighted ℓ_1 terms can be merged. Define the combined nonnegative weight:

$$\boldsymbol{C}_t = \lambda \boldsymbol{W}_t + \gamma(1-\boldsymbol{M}_t), \quad (8)$$

so that $\lambda\|\boldsymbol{W}_t \odot \boldsymbol{S}_t\|_1 + \gamma\|(1-\boldsymbol{M}_t)\odot \boldsymbol{S}_t\|_1 = \|\boldsymbol{C}_t \odot \boldsymbol{S}_t\|_1$.

For each incoming frame, we solve problem (3) using an efficient alternating direction method of multipliers (ADMM) procedure [22], [23]. The augmented Lagrangian function is:

Algorithm 1 MC-RPCA

Input: IR sequence $\{\boldsymbol{I}_t\}_{t=1}^T$; parameters $\lambda, \gamma, \varepsilon$; weight-mapping parameters (τ, k, w_l, w_h); ADMM parameters $\rho, \mu_0, \mu_{\max}$; decay scale σ_m.
Output: Per-frame decomposition $\{(\boldsymbol{L}_t, \boldsymbol{S}_t)\}_{t=1}^T$.
Initialize $\boldsymbol{S}_0 \leftarrow 0$.
for $t = 1$ to T **do**
  Estimate optical flow from $(\boldsymbol{I}_{t-1}, \boldsymbol{I}_t)$ and remove global camera motion.
  Construct motion weights $\boldsymbol{W}_t$ using (4).
  if $t = 1$ **then**
    Set $\boldsymbol{M}_1 \leftarrow 0$ {No historical trajectory prior}
  else
    Target prediction: $\boldsymbol{P}_t \leftarrow \boldsymbol{T}_{t-1\to t}(\boldsymbol{S}_{t-1})$.
    Compute trajectory-confidence map $\boldsymbol{M}_t$ using (7).
  end if
  Compute combined weights $\boldsymbol{C}_t \leftarrow \lambda \boldsymbol{W}_t + \gamma(1-\boldsymbol{M}_t)$.
  Initialize inner loop: $\boldsymbol{L}_t^0 \leftarrow \boldsymbol{0}$, $\boldsymbol{S}_t^0 \leftarrow \boldsymbol{0}$, $\boldsymbol{Y}_t^0 \leftarrow \boldsymbol{0}$, $\mu \leftarrow \mu_0$.
  for $k = 0$ to $K_{\max} - 1$ **do**
    $\boldsymbol{Q}_t^k \leftarrow \boldsymbol{I}_t - \boldsymbol{S}_t^k + \mu^{-1}\boldsymbol{Y}_t^k$.
    $\boldsymbol{L}_t^{k+1} \leftarrow \mathrm{SVT}_{1/\mu}(\boldsymbol{Q}_t^k)$.
    $\boldsymbol{R}_t^k \leftarrow \boldsymbol{I}_t - \boldsymbol{L}_t^{k+1} + \mu^{-1}\boldsymbol{Y}_t^k$.
    $\boldsymbol{S}_t^{k+1} \leftarrow \mathrm{sign}(\boldsymbol{R}_t^k)\odot \max\Big(|\boldsymbol{R}_t^k| - \frac{\boldsymbol{C}_t}{\mu}, 0\Big)$.
    $\boldsymbol{Y}_t^{k+1} \leftarrow \boldsymbol{Y}_t^k + \mu(\boldsymbol{I}_t - \boldsymbol{L}_t^{k+1} - \boldsymbol{S}_t^{k+1})$.
    $\mu \leftarrow \min(\rho\mu, \mu_{\max})$.
    if $\|\boldsymbol{I}_t - \boldsymbol{L}_t^{k+1} - \boldsymbol{S}_t^{k+1}\|_F / \|\boldsymbol{I}_t\|_F < \varepsilon$ **then**
      break
    end if
  end for
  Output $(\boldsymbol{L}_t, \boldsymbol{S}_t) \leftarrow (\boldsymbol{L}_t^{k+1}, \boldsymbol{S}_t^{k+1})$ and store $\boldsymbol{S}_t$ to construct $\boldsymbol{M}_{t+1}$.
end for

$$\mathcal{L}(\boldsymbol{L}_t, \boldsymbol{S}_t, \boldsymbol{Y}_t) = \|\boldsymbol{L}_t\|_* + \|\boldsymbol{C}_t \odot \boldsymbol{S}_t\|_1 + \langle \boldsymbol{Y}_t,\ \boldsymbol{I}_t - \boldsymbol{L}_t - \boldsymbol{S}_t\rangle + \frac{\mu}{2}\|\boldsymbol{I}_t - \boldsymbol{L}_t - \boldsymbol{S}_t\|_F^2, \quad (9)$$

where $\boldsymbol{Y}_t$ is the dual multiplier and $\mu > 0$ is a penalty parameter.

At inner iteration k, the updates are:

- *Low-rank update (SVT).* Let

$$\boldsymbol{Q}_t^k = \boldsymbol{I}_t - \boldsymbol{S}_t^k + \mu^{-1}\boldsymbol{Y}_t^k, \quad (10)$$

$$\boldsymbol{L}_t^{k+1} = \mathrm{SVT}_{1/\mu}(\boldsymbol{Q}_t^k), \quad (11)$$

  where $\mathrm{SVT}_\tau(\cdot)$ is the singular value thresholding operator with threshold $\tau > 0$. Given $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, the singular value thresholding operator is

$$\mathrm{SVT}_\tau(\boldsymbol{X}) = \boldsymbol{U}\,\mathrm{diag}\big(\max(\sigma_i - \tau, 0)\big)\,\boldsymbol{V}^\top. \quad (12)$$

- *Sparse update (pixel-wise weighted shrinkage).* Let

$$\boldsymbol{R}_t^k = \boldsymbol{I}_t - \boldsymbol{L}_t^{k+1} + \mu^{-1}\boldsymbol{Y}_t^k, \quad (13)$$

$$\boldsymbol{S}_t^{k+1} = \mathrm{sign}(\boldsymbol{R}_t^k)\odot \max\left(|\boldsymbol{R}_t^k| - \frac{\boldsymbol{C}_t}{\mu},\ 0\right). \quad (14)$$

- *Multiplier and penalty update.*

$$\boldsymbol{Y}_t^{k+1} = \boldsymbol{Y}_t^k + \mu\left(\boldsymbol{I}_t - \boldsymbol{L}_t^{k+1} - \boldsymbol{S}_t^{k+1}\right). \quad (15)$$

  Optionally, μ can be increased adaptively as $\mu \leftarrow \min(\rho\mu, \mu_{\max})$ with $\rho > 1$.

After convergence, we output $(\boldsymbol{L}_t, \boldsymbol{S}_t)$ for the current frame and store $\boldsymbol{S}_t$ to construct $\boldsymbol{M}_{t+1}$ in the next time step. Algorithm 1 summarizes the overall streaming processing.

V. Experiments

A. Experimental Setup

We evaluate our method on the WideIRSTD dataset [24]. In IR imagery, both the background and the target may be in motion; however, they typically move at different velocities. In our experiments, we set $\lambda = \frac{1}{\sqrt{\max(m,n)}}$ and $\gamma = 0.6$. The motion weight mapping parameters (τ, k, w_l, w_h) are set as $\tau = Quantile_{0.95}(\boldsymbol{V}_t^{com})$, $k = 0.5 \cdot \tau$, $w_l = 0.2$, and $w_h = 5$. For the trajectory-confidence map, we set the decay scale $\sigma_m = 2$ to allow for moderate spatial uncertainty.

B. Comparison Results

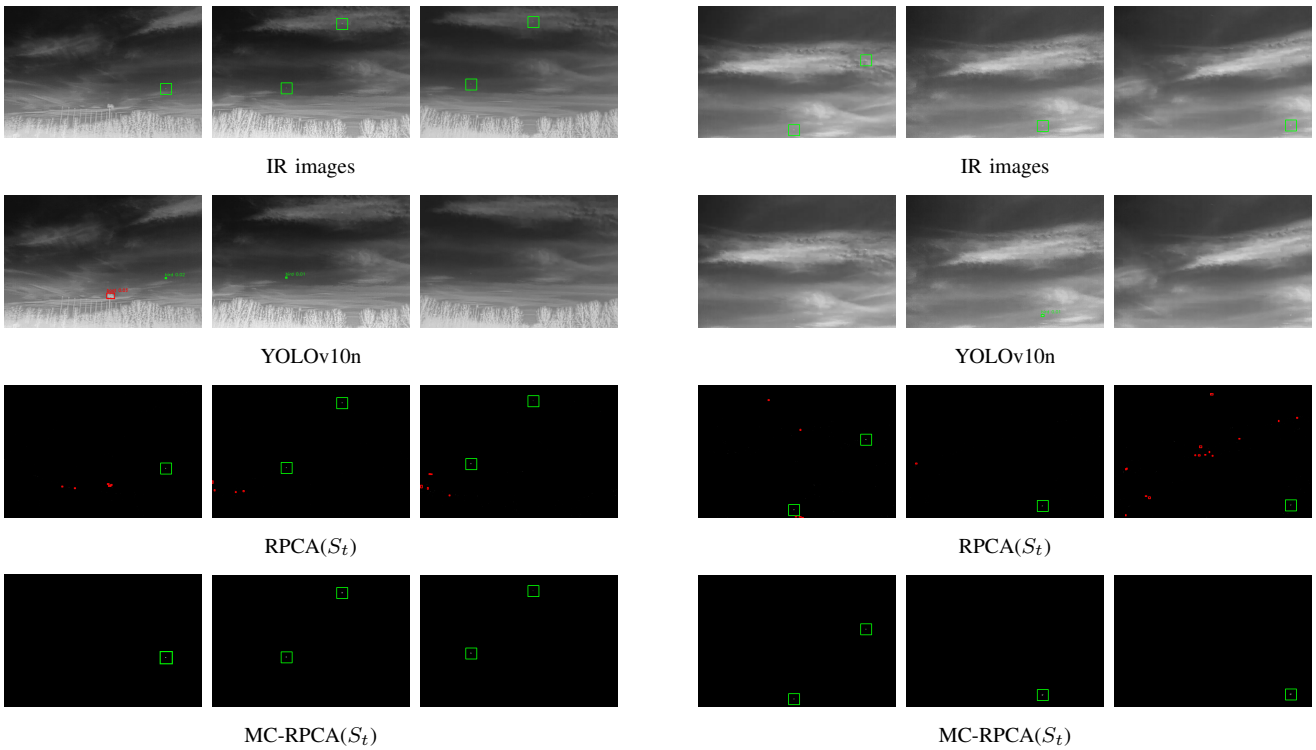
We compare our method MC-RPCA with classic RPCA and a state-of-the-art deep detector YOLOv10n [15]. Figure 1 illustrates the detection results of YOLOv10n in two typical scenarios, as well as the target component S_t separated by RPCA and MC-RPCA. The green boxes indicate true targets, while the red boxes denote false alarms. In both scenarios, classic RPCA yields frequent false alarms, which come from the background interference. Due to the lack of high-quality datasets and its limited sensitivity to small target characteristics, YOLO often suffers from missed detections. In contrast, the proposed MC-RPCA effectively suppresses background-induced artifacts by leveraging motion-derived weights and trajectory continuity, resulting in cleaner sparse maps with fewer false positives and more stable target detections.

Table I reports the frame-level false alarm rate and miss rate averaged over 145 frames. Specifically, the false alarm rate is defined as the proportion of frames that contain at least one false alarm, and the miss rate is defined as the proportion of frames where the target is missed. These results demonstrate that our method MC-RPCA yields a significantly lower false alarm rate than YOLOv10n and RPCA, while achieving a low miss rate, which benefits from exploiting spatio-temporal target characteristics via motion constraints.

TABLE I
False alarm and miss rate comparison.

Method	False alarm rate	Miss rate
RPCA	0.46	0.05
YOLOv10n	0.54	0.55
MC-RPCA	**0.15**	**0.05**

In conclusion, we propose an MC-RPCA framework for IR small moving target detection by integrating motion-derived sparsity reweighting and a causal trajectory-confidence prior. The resulting per-frame convex problem is efficiently solved via ADMM, and experiments demonstrate fewer false alarms and more stable detections than classic RPCA.



(a) Scenario 1 (b) Scenario 2

Fig. 1. Detection results of three different methods under two typical scenarios.

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